Laws of Surds

Worked Solutions with TikZ Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref `a5`)

This document contains all 40 *Laws of Surds* questions, each with a fully worked solution and a TikZ diagram matched to the task:

- **Factor trees** for simplifying $\sqrt{n}$ (split off the largest perfect square);
- **Surd-law cards** for multiplying, squaring and fraction surds;
- **Area-model (box) grids** for expanding brackets, difference of two squares and perfect squares;
- **Solution-flow** diagrams for adding/subtracting surds and fraction products;
- a **right-triangle** diagram for the Pythagoras problem.

Contents

Laws of Surds 01 (a5-001) 1

Laws of Surds 02 (a5-002) 2

Laws of Surds 03 (a5-003) 2

Laws of Surds 04 (a5-004) 3

Laws of Surds 05 (a5-005) 3

Laws of Surds 06 (a5-006) 4

Laws of Surds 07 (a5-007) 5

Laws of Surds 08 (a5-008) 5

Laws of Surds 09 (a5-009) 6

Laws of Surds 10 (a5-010) 7

Laws of Surds 11 (a5-011) 7

Laws of Surds 12 (a5-012) 8

Laws of Surds 13 (a5-013) 8

Laws of Surds 14 (a5-014) 9

Laws of Surds 15 (a5-015) 9

Laws of Surds 16 (a5-016) 10

Laws of Surds 17 (a5-017) 10

Laws of Surds 18 (a5-018) 11

Laws of Surds 19 (a5-019) 11

Laws of Surds 20 (a5-020) 12

Laws of Surds 21 (a5-021) 12

Laws of Surds 22 (a5-022) 13

Laws of Surds 23 (a5-023) 13

Laws of Surds 24 (a5-024) 14

Laws of Surds 25 (a5-025) 14

Laws of Surds 26 (a5-026) 15

Laws of Surds 27 (a5-027) 15

Laws of Surds 28 (a5-028) 16

Laws of Surds 29 (a5-029) 16

Laws of Surds 30 (a5-030) 17

Laws of Surds 31 (a5-031) 18

Laws of Surds 32 (a5-032) 18

Laws of Surds 33 (a5-033) 19

Laws of Surds 34 (a5-034) 20

Laws of Surds 35 (a5-035) 20

Laws of Surds 36 (a5-036) 21

Laws of Surds 37 (a5-037) 22

Laws of Surds 38 (a5-038) **22**

Laws of Surds 39 (a5-039) **23**

Laws of Surds 40 (a5-040) **23**

Laws of Surds 01 (a5-001)

Difficulty: Foundation | Marks: 1

Simplify: $\sqrt{12}$

Worked solution.

Step 1: Find the largest square number that is a factor of 12.

$$\sqrt{12} = \sqrt{4 \times 3}$$

4 is the largest square number that divides into 12.

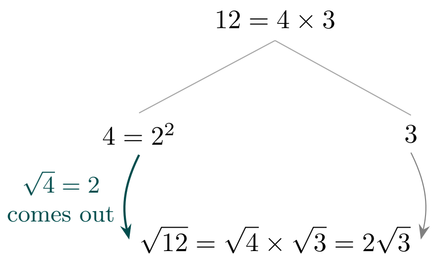
Step 2: Split using the rule $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ *and simplify.*

$$\sqrt{4} \times \sqrt{3} = 2\sqrt{3}$$

$\sqrt{4} = 2$, so the surd simplifies to $2\sqrt{3}$.

Final answer: $2\sqrt{3}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 02 (a5-002)

Difficulty: Foundation | Marks: 1

Simplify: $\sqrt{18}$

Worked solution.

Step 1: Find the largest square number that is a factor of 18.

$$\sqrt{18} = \sqrt{9 \times 2}$$

9 is the largest square number that divides into 18.

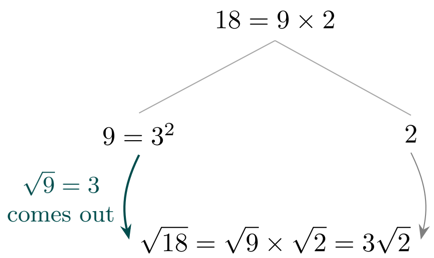
Step 2: Split and simplify.

$$\sqrt{9} \times \sqrt{2} = 3\sqrt{2}$$

$\sqrt{9} = 3$.

Final answer: $3\sqrt{2}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 03 (a5-003)

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{45}$

Worked solution.

Step 1: Find the largest square number that is a factor of 45.

$$\sqrt{45} = \sqrt{9 \times 5}$$

9 is the largest square factor of 45.

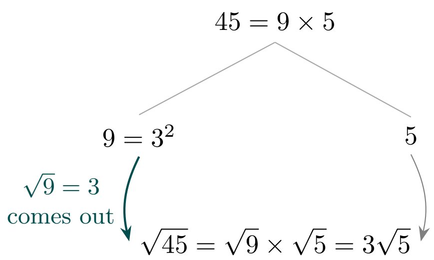
Step 2: Split and simplify.

$$\sqrt{9} \times \sqrt{5} = 3\sqrt{5}$$

$\sqrt{9} = 3$.

Final answer: $3\sqrt{5}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 04 `(a5-004)`

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{75}$

Worked solution.

Step 1: Find the largest square number that is a factor of 75.

$$\sqrt{75} = \sqrt{25 \times 3}$$

25 is the largest square factor of 75.

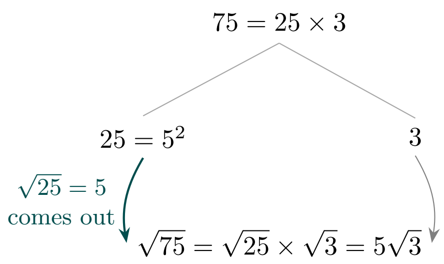
Step 2: Split and simplify.

$$\sqrt{25} \times \sqrt{3} = 5\sqrt{3}$$

$\sqrt{25} = 5$.

Final answer: $5\sqrt{3}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 05 `(a5-005)`

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{98}$

Worked solution.

Step 1: Find the largest square number that is a factor of 98.

$$\sqrt{98} = \sqrt{49 \times 2}$$

49 is the largest square factor of 98.

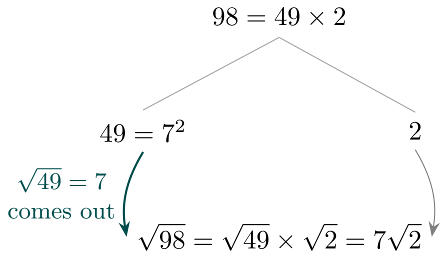
Step 2: Split and simplify.

$$\sqrt{49} \times \sqrt{2} = 7\sqrt{2}$$

$\sqrt{49} = 7$.

Final answer: $7\sqrt{2}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 06 `(a5-006)`

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{\frac{7}{4}}$

Worked solution.

Step 1: Use the rule $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$.

$$\frac{\sqrt{7}}{\sqrt{4}}$$

Split the surd across the numerator and denominator.

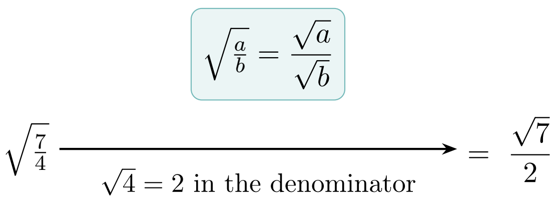
Step 2: Simplify the denominator.

$$\frac{\sqrt{7}}{2}$$

$\sqrt{4} = 2$. The numerator $\sqrt{7}$ cannot be simplified further.

Final answer: $\frac{\sqrt{7}}{2}$

Diagram.



Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 07 `(a5-007)`

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{\frac{9}{25}}$

Worked solution.

Step 1: Use the rule $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$.

$$\frac{\sqrt{9}}{\sqrt{25}}$$

Split the surd across the fraction.

Step 2: Evaluate both square roots.

$$\frac{3}{5}$$

$\sqrt{9} = 3$ and $\sqrt{25} = 5$. Both are perfect squares, so the answer is rational.

Final answer: $\frac{3}{5}$

Diagram.

$$\boxed{\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}}$$

$$\sqrt{\frac{9}{25}} \xrightarrow[\sqrt{9}=3,\ \sqrt{25}=5]{} = \frac{3}{5}$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 08 (a5-008)

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{\frac{11}{16}}$

Worked solution.

Step 1: Split the surd across the fraction.

$$\frac{\sqrt{11}}{\sqrt{16}}$$

$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$.

Step 2: Simplify the denominator.

$$\frac{\sqrt{11}}{4}$$

$\sqrt{16} = 4$. The numerator stays as $\sqrt{11}$.

Final answer: $\frac{\sqrt{11}}{4}$

Diagram.

$$\boxed{\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}}$$

$$\sqrt{\frac{11}{16}} \xrightarrow[\sqrt{16}=4]{} = \frac{\sqrt{11}}{4}$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 09 (a5-009)

Difficulty: Foundation | Marks: 2

Evaluate: $3\sqrt{2} \times 5\sqrt{2}$. Give your answer as a whole number.

Worked solution.

Step 1: Multiply the whole-number parts together and the surd parts together.

$$(3 \times 5) \times (\sqrt{2} \times \sqrt{2}) = 15 \times \sqrt{4}$$

Use the rule $\sqrt{a} \times \sqrt{a} = a$.

Step 2: Simplify.

$$15 \times 2 = 30$$

$\sqrt{2} \times \sqrt{2} = 2$, so the surds disappear entirely.

Final answer: 30

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{a} = mn\,a}$$

$$3\sqrt{2} \times 5\sqrt{2} \xrightarrow[3 \times 5 = 15,\ \sqrt{2} \times \sqrt{2} = 2,\ \text{so } 15 \times 2]{} = \ 30$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 10 (a5-010)

Difficulty: Foundation | Marks: 1

Evaluate: $\sqrt{6} \times \sqrt{6}$

Worked solution.

Apply the key rule $(\sqrt{a})^2 = a$.

$$\sqrt{6} \times \sqrt{6} = 6$$

Squaring a surd removes the root sign. This follows directly from the definition of a square root.

Final answer: 6

Diagram.

$$\boxed{\sqrt{a} \times \sqrt{a} = a}$$

$$\sqrt{6} \times \sqrt{6} \xrightarrow[\text{a root times itself removes the root}]{} = \ 6$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 11 `(a5-011)`

Difficulty: Foundation | Marks: 1

Evaluate: $\sqrt{3} \times 4\sqrt{3}$

Worked solution.

Multiply the surd parts and the number parts.

$$1 \times 4 \times (\sqrt{3} \times \sqrt{3}) = 4 \times 3 = 12$$

$\sqrt{3} \times \sqrt{3} = 3$, then multiply by 4.

Final answer: 12

Diagram.

$$\boxed{\sqrt{a} \times m\sqrt{a} = m\,a}$$

$$\sqrt{3} \times 4\sqrt{3} \xrightarrow[4 \times (\sqrt{3})^2 = 4 \times 3]{} = \; 12$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 12 `(a5-012)`

Difficulty: Foundation | Marks: 2

Evaluate: $2\sqrt{3} \times 4\sqrt{7}$**. Give your answer as a surd.**

Worked solution.

Step 1: Multiply the whole-number parts together and the surd parts together.

$$(2 \times 4) \times (\sqrt{3} \times \sqrt{7})$$

Deal with the numbers and the surds separately.

Step 2: Simplify using $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$.

$$8\sqrt{21}$$

$2 \times 4 = 8$ and $\sqrt{3} \times \sqrt{7} = \sqrt{21}$. Since 21 has no square factors, this is fully simplified.

Final answer: $8\sqrt{21}$

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{b} = mn\sqrt{ab}}$$

$$2\sqrt{3} \times 4\sqrt{7} \xrightarrow[2 \times 4 = 8,\ \sqrt{3} \times \sqrt{7} = \sqrt{21}]{} = \; 8\sqrt{21}$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 13 `(a5-013)`

Difficulty: Foundation | *Marks: 1*

Evaluate: $(\sqrt{5})^2$

Worked solution.

Squaring a square root cancels out the root.

$$(\sqrt{5})^2 = 5$$

By definition, $(\sqrt{a})^2 = a$. Squaring undoes the square root.

Final answer: 5

Diagram.

$$\boxed{(\sqrt{a})^2 = a}$$

$$(\sqrt{5})^2 \xrightarrow[\text{squaring undoes the square root}]{} = 5$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 14 `(a5-014)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $(3\sqrt{7})^2$

Worked solution.

Step 1: Square both the number part and the surd part.

$$3^2 \times (\sqrt{7})^2 = 9 \times 7$$

$(a\sqrt{b})^2 = a^2 \times b$.

Step 2: Evaluate.

$$63$$

$9 \times 7 = 63$.

Final answer: 63

Diagram.

$$\boxed{(m\sqrt{a})^2 = m^2 a}$$

$$(3\sqrt{7})^2 \xrightarrow[3^2 \times 7 = 9 \times 7]{} = 63$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 15 (a5-015)

Difficulty: Foundation | Marks: 2

Evaluate: $(2\sqrt{13})^2$

Worked solution.

Step 1: Square the coefficient and the surd separately.

$$2^2 \times (\sqrt{13})^2 = 4 \times 13$$

The square of 2 is 4, and squaring $\sqrt{13}$ gives 13.

Step 2: Evaluate.

$$52$$

$4 \times 13 = 52$.

Final answer: 52

Diagram.

$$\boxed{(m\sqrt{a})^2 = m^2 a}$$

$$(2\sqrt{13})^2 \xrightarrow[2^2 \times 13 = 4 \times 13]{\hspace{10em}} = \ 52$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 16 (a5-016)

Difficulty: Foundation | Marks: 2

Evaluate: $3\sqrt{6} \times 2\sqrt{6}$

Worked solution.

Step 1: Multiply the coefficients and the surds separately.

$$(3 \times 2) \times (\sqrt{6} \times \sqrt{6}) = 6 \times 6$$

$\sqrt{6} \times \sqrt{6} = 6$.

Step 2: Evaluate.

$$36$$

$6 \times 6 = 36$.

Final answer: 36

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{a} = mn\, a}$$

$$3\sqrt{6} \times 2\sqrt{6} \xrightarrow[3 \times 2 \times 6]{\hspace{10em}} = \ 36$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 17 `(a5-017)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $2\sqrt{3} \times 5\sqrt{27}$**. Give your answer as a whole number.**

Worked solution.

Step 1: Multiply the coefficients and the surds separately.

$$(2 \times 5) \times (\sqrt{3} \times \sqrt{27}) = 10 \times \sqrt{81}$$

$\sqrt{3} \times \sqrt{27} = \sqrt{3 \times 27} = \sqrt{81}$.

Step 2: Evaluate.

$$10 \times 9 = 90$$

$\sqrt{81} = 9$.

Final answer: 90

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{b} = mn\sqrt{ab}}$$

$$2\sqrt{3} \times 5\sqrt{27} \xrightarrow[10\sqrt{81} = 10 \times 9]{} = 90$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 18 `(a5-018)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $3\sqrt{5} \times 2\sqrt{45}$**. Give your answer as a whole number.**

Worked solution.

Step 1: Multiply the coefficients and the surds separately.

$$(3 \times 2) \times (\sqrt{5} \times \sqrt{45}) = 6 \times \sqrt{225}$$

$\sqrt{5} \times \sqrt{45} = \sqrt{5 \times 45} = \sqrt{225}$.

Step 2: Evaluate.

$$6 \times 15 = 90$$

$\sqrt{225} = 15$.

Final answer: 90

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{b} = mn\sqrt{ab}}$$

$$3\sqrt{5} \times 2\sqrt{45} \xrightarrow[6\sqrt{225} = 6 \times 15]{} = 90$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 19 `(a5-019)`

Difficulty: Foundation | Marks: 2

Evaluate: $\frac{\sqrt{10}}{6} \times \frac{12}{\sqrt{5}}$

Worked solution.

Step 1: Multiply the fractions together.

$$\frac{12\sqrt{10}}{6\sqrt{5}}$$

Multiply numerators and denominators: $\frac{\sqrt{10}\times 12}{6\times\sqrt{5}}$.

Step 2: Simplify the number part and the surd part separately.

$$\frac{12}{6} \times \frac{\sqrt{10}}{\sqrt{5}} = 2 \times \sqrt{\frac{10}{5}} = 2\sqrt{2}$$

$\frac{12}{6} = 2$ and $\frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2}$.

Final answer: $2\sqrt{2}$

Diagram.

$$\frac{\sqrt{10}}{6} \times \frac{12}{\sqrt{5}} \xrightarrow{\text{multiply across}} \frac{12\sqrt{10}}{6\sqrt{5}} \xrightarrow{\text{simplify}} 2\sqrt{\tfrac{10}{5}} = 2\sqrt{2}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 20 `(a5-020)`

Difficulty: Foundation | Marks: 2

Evaluate: $\frac{\sqrt{12}}{3} \times \frac{2}{\sqrt{27}}$

Worked solution.

Step 1: Multiply the fractions together.

$$\frac{2\sqrt{12}}{3\sqrt{27}}$$

Multiply the numerators and the denominators.

Step 2: Simplify the surd part.

$$\frac{2}{3} \times \sqrt{\frac{12}{27}} = \frac{2}{3} \times \sqrt{\frac{4}{9}} = \frac{2}{3} \times \frac{2}{3}$$

$\frac{12}{27} = \frac{4}{9}$, and $\sqrt{\frac{4}{9}} = \frac{2}{3}$.

Step 3: Evaluate.

$$\frac{4}{9}$$

$\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$.

Final answer: $\frac{4}{9}$

Diagram.

$$\frac{\sqrt{12}}{3} \times \frac{2}{\sqrt{27}} \xrightarrow{\text{multiply across}} \frac{2\sqrt{12}}{3\sqrt{27}} = \frac{4\sqrt{3}}{9\sqrt{3}} \xrightarrow{\text{simplify}} \frac{4}{9}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 21 `(a5-021)`

Difficulty: Foundation | Marks: 2

Express $\sqrt{48} + \sqrt{27}$ in the form $k\sqrt{3}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd by finding the largest square factor.

$$\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3} \quad \text{and} \quad \sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$$

16 is the largest square factor of 48, and 9 is the largest square factor of 27.

Step 2: Add the like surds.

$$4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$$

These are like surds (both $\sqrt{3}$), so add the coefficients: $4 + 3 = 7$.

Final answer: $7\sqrt{3}$

Diagram.

$$\sqrt{48} + \sqrt{27} \xrightarrow{\text{simplify each surd}} 4\sqrt{3} + 3\sqrt{3} \xrightarrow{\text{collect like surds}} 7\sqrt{3}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 22 `(a5-022)`

Difficulty: Foundation | Marks: 2

Express $\sqrt{50} - \sqrt{8}$ in the form $k\sqrt{2}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd.

$$\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2} \quad \text{and} \quad \sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}$$

25 is the largest square factor of 50, and 4 is the largest square factor of 8.

Step 2: Subtract the like surds.

$$5\sqrt{2} - 2\sqrt{2} = 3\sqrt{2}$$

Subtract the coefficients: $5 - 2 = 3$.

Final answer: $3\sqrt{2}$

Diagram.

$$\sqrt{50}-\sqrt{8} \xrightarrow{\text{simplify each surd}} 5\sqrt{2}-2\sqrt{2} \xrightarrow{\text{collect like surds}} 3\sqrt{2}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 23 `(a5-023)`

Difficulty: Foundation | Marks: 2

Express $3\sqrt{12}-2\sqrt{75}$ in the form $k\sqrt{3}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd.

$$3\sqrt{12} = 3\times 2\sqrt{3} = 6\sqrt{3} \quad \text{and} \quad 2\sqrt{75} = 2\times 5\sqrt{3} = 10\sqrt{3}$$

$\sqrt{12} = 2\sqrt{3}$ and $\sqrt{75} = 5\sqrt{3}$. Then multiply by the coefficients outside.

Step 2: Subtract the like surds.

$$6\sqrt{3}-10\sqrt{3} = -4\sqrt{3}$$

$6-10=-4$. A negative answer is perfectly valid here.

Final answer: $-4\sqrt{3}$

Diagram.

$$3\sqrt{12}-2\sqrt{75} \xrightarrow{\text{simplify each surd}} 6\sqrt{3}-10\sqrt{3} \xrightarrow{\text{collect like surds}} -4\sqrt{3}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 24 `(a5-024)`

Difficulty: Foundation | Marks: 2

Express $2\sqrt{18}+3\sqrt{50}$ in the form $k\sqrt{2}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd.

$$2\sqrt{18} = 2\times 3\sqrt{2} = 6\sqrt{2} \quad \text{and} \quad 3\sqrt{50} = 3\times 5\sqrt{2} = 15\sqrt{2}$$

$\sqrt{18} = 3\sqrt{2}$ and $\sqrt{50} = 5\sqrt{2}$.

Step 2: Add the like surds.

$$6\sqrt{2}+15\sqrt{2} = 21\sqrt{2}$$

$6+15=21$.

Final answer: $21\sqrt{2}$

Diagram.

$$2\sqrt{18} + 3\sqrt{50} \xrightarrow{\text{simplify each surd}} 6\sqrt{2} + 15\sqrt{2} \xrightarrow{\text{collect like surds}} 21\sqrt{2}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 25 (a5-025)

Difficulty: Foundation | Marks: 3

Express $5\sqrt{28} + 3\sqrt{63} - 2\sqrt{7}$ in the form $k\sqrt{7}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd.

$$5\sqrt{28} = 5 \times 2\sqrt{7} = 10\sqrt{7} \quad \text{and} \quad 3\sqrt{63} = 3 \times 3\sqrt{7} = 9\sqrt{7}$$

$\sqrt{28} = 2\sqrt{7}$ and $\sqrt{63} = 3\sqrt{7}$. The third term is already in surd form.

Step 2: Combine the like surds.

$$10\sqrt{7} + 9\sqrt{7} - 2\sqrt{7} = 17\sqrt{7}$$

$10 + 9 - 2 = 17$.

Final answer: $17\sqrt{7}$

Diagram.

$$5\sqrt{28} + 3\sqrt{63} - 2\sqrt{7} \xrightarrow{\text{simplify each surd}} 10\sqrt{7} + 9\sqrt{7} - 2\sqrt{7} \xrightarrow{\text{collect}} 17\sqrt{7}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 26 (a5-026)

Difficulty: Foundation | Marks: 3

Express $2\sqrt{20} + 3\sqrt{45} + \sqrt{80}$ in the form $k\sqrt{5}$, where k is an integer.

Worked solution.

Step 1: Simplify each surd.

$$2\sqrt{20} = 2 \times 2\sqrt{5} = 4\sqrt{5}, \quad 3\sqrt{45} = 3 \times 3\sqrt{5} = 9\sqrt{5}, \quad \sqrt{80} = 4\sqrt{5}$$

$\sqrt{20} = 2\sqrt{5}$, $\sqrt{45} = 3\sqrt{5}$, and $\sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5}$.

Step 2: Add the like surds.

$$4\sqrt{5} + 9\sqrt{5} + 4\sqrt{5} = 17\sqrt{5}$$

$4 + 9 + 4 = 17$.

Final answer: $17\sqrt{5}$

Diagram.

$$2\sqrt{20}+3\sqrt{45}+\sqrt{80} \xrightarrow{\text{simplify each surd}} 4\sqrt{5}+9\sqrt{5}+4\sqrt{5} \xrightarrow{\text{collect}} 17\sqrt{5}$$

Solution flow: each arrow applies one surd law.

Laws of Surds 27 `(a5-027)`

Difficulty: Foundation | Marks: 3

Expand and simplify: $(2+\sqrt{3})(4+\sqrt{3})$

Worked solution.

Expand and simplify

$$\begin{aligned}(2+\sqrt{3})(4+\sqrt{3}) &= 2(4+\sqrt{3})+\sqrt{3}(4+\sqrt{3})\\ &= 8+2\sqrt{3}+4\sqrt{3}+3\\ &= 11+6\sqrt{3}\end{aligned}$$

Distribute each term in the first bracket across the second. Then collect rational parts $(8+3=11)$ and surd parts $(2\sqrt{3}+4\sqrt{3}=6\sqrt{3})$.

Final answer: $11+6\sqrt{3}$

Diagram.

$\times$	4	$\sqrt{3}$
2	8	$2\sqrt{3}$
$\sqrt{3}$	$4\sqrt{3}$	3

Collect: rational $8+3=11$; surd $2\sqrt{3}+4\sqrt{3}=6\sqrt{3}$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 28 `(a5-028)`

Difficulty: Foundation | Marks: 3

Expand and simplify: $(5+2\sqrt{7})(3-\sqrt{7})$

Worked solution.

Expand and simplify

$$\begin{aligned}(5+2\sqrt{7})(3-\sqrt{7}) &= 5(3-\sqrt{7})+2\sqrt{7}(3-\sqrt{7})\\ &= 15-5\sqrt{7}+6\sqrt{7}-2(7)\\ &= 15-14-5\sqrt{7}+6\sqrt{7}\\ &= 1+\sqrt{7}\end{aligned}$$

Distribute each term. Note $2\sqrt{7}\times\sqrt{7}=2\times 7=14$. Then collect rational and surd parts.

Final answer: $1+\sqrt{7}$

Diagram.

$\times$	3	$-\sqrt{7}$
5	15	$-5\sqrt{7}$
$2\sqrt{7}$	$6\sqrt{7}$	-14

Collect: $15-14=1$; $-5\sqrt{7}+6\sqrt{7}=\sqrt{7}$. Note $2\sqrt{7}\times(-\sqrt{7})=-2\times 7=-14$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 29 (a5-029)

Difficulty: Foundation | Marks: 3

Expand and simplify: $(\sqrt{5}+3)(\sqrt{5}-1)$

Worked solution.

Expand and simplify

$$\begin{aligned}(\sqrt{5}+3)(\sqrt{5}-1) &= \sqrt{5}(\sqrt{5}-1)+3(\sqrt{5}-1)\\ &= 5-\sqrt{5}+3\sqrt{5}-3\\ &= 2+2\sqrt{5}\end{aligned}$$

Distribute each term. Rational parts: $5-3=2$. Surd parts: $-\sqrt{5}+3\sqrt{5}=2\sqrt{5}$.

Final answer: $2+2\sqrt{5}$

Diagram.

$\times$	$\sqrt{5}$	-1
$\sqrt{5}$	5	$-\sqrt{5}$
3	$3\sqrt{5}$	-3

Collect: $5-3=2$; $-\sqrt{5}+3\sqrt{5}=2\sqrt{5}$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 30 `(a5-030)`

Difficulty: Foundation | *Marks: 2*

Expand and simplify: $(\sqrt{11}+3)(\sqrt{11}-3)$

Worked solution.

Step 1: Recognise this as the difference of two squares pattern: $(\sqrt{a}+b)(\sqrt{a}-b)=a-b^2$.

$$(\sqrt{11})^2-3^2$$

The two brackets are conjugates — one has a plus, the other has a minus — so the middle surd terms cancel.

Step 2: Evaluate.

$$11-9=2$$

The result is a rational number with no surds remaining. This is the key idea behind rationalising denominators.

Final answer: 2

Diagram.

$\times$	$\sqrt{11}$	-3
$\sqrt{11}$	11	$-3\sqrt{11}$
3	$3\sqrt{11}$	-9

Difference of two squares: the surd cross-terms cancel, leaving $11-9=2$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 31 `(a5-031)`

Difficulty: Foundation | *Marks: 2*

Expand and simplify: $(7-2\sqrt{3})(7+2\sqrt{3})$

Worked solution.

Step 1: Apply the difference of two squares: $(a-b)(a+b)=a^2-b^2$.

$$7^2-(2\sqrt{3})^2$$

These are conjugate pairs, so the cross terms cancel.

Step 2: Evaluate each square.

$$49-4\times 3=49-12=37$$

$(2\sqrt{3})^2=4\times 3=12$.

Final answer: 37

Diagram.

$\times$	7	$2\sqrt{3}$
7	49	$14\sqrt{3}$
$-2\sqrt{3}$	$-14\sqrt{3}$	-12

DOTS: cross-terms cancel; $(2\sqrt{3})^2 = 4\times 3 = 12$, so $49 - 12 = 37$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 32 `(a5-032)`

Difficulty: Foundation | *Marks: 3*

Expand and simplify: $(\sqrt{3}+2)^2$

Worked solution.

Expand and simplify

$$\begin{aligned}(\sqrt{3}+2)^2 &= (\sqrt{3}+2)(\sqrt{3}+2)\\ &= \sqrt{3}(\sqrt{3}+2)+2(\sqrt{3}+2)\\ &= 3+2\sqrt{3}+2\sqrt{3}+4\\ &= 7+4\sqrt{3}\end{aligned}$$

Write the square as two identical brackets, then distribute each term.

Final answer: $7+4\sqrt{3}$

Diagram.

$\times$	$\sqrt{3}$	2
$\sqrt{3}$	3	$2\sqrt{3}$
2	$2\sqrt{3}$	4

Perfect square: $3+4=7$ and $2\sqrt{3}+2\sqrt{3}=4\sqrt{3}$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 33 `(a5-033)`

Difficulty: Foundation | *Marks: 3*

Expand and simplify: $(2\sqrt{5}-3)^2$

Worked solution.

Expand and simplify

$$\begin{aligned}(2\sqrt{5}-3)^2 &= (2\sqrt{5}-3)(2\sqrt{5}-3)\\ &= 2\sqrt{5}(2\sqrt{5}-3)-3(2\sqrt{5}-3)\\ &= 4(5)-6\sqrt{5}-6\sqrt{5}+9\\ &= 20-12\sqrt{5}+9\\ &= 29-12\sqrt{5}\end{aligned}$$

Write the square as two identical brackets, then distribute each term. Note $2\sqrt{5}\times 2\sqrt{5}=4\times 5=20$.

Final answer: $29-12\sqrt{5}$

Diagram.

$\times$	$2\sqrt{5}$	-3
$2\sqrt{5}$	20	$-6\sqrt{5}$
-3	$-6\sqrt{5}$	9

Perfect square: $(2\sqrt{5})^2=4\times 5=20$; $20+9=29$, $-6\sqrt{5}-6\sqrt{5}=-12\sqrt{5}$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 34 `(a5-034)`

Difficulty: Foundation | Marks: 3

Expand and simplify: $(4+3\sqrt{2})(1-2\sqrt{2})$

Worked solution.

Expand and simplify

$$\begin{aligned}(4+3\sqrt{2})(1-2\sqrt{2}) &= 4(1-2\sqrt{2})+3\sqrt{2}(1-2\sqrt{2})\\ &= 4-8\sqrt{2}+3\sqrt{2}-6(2)\\ &= 4-8\sqrt{2}+3\sqrt{2}-12\\ &= -8-5\sqrt{2}\end{aligned}$$

Distribute each term. Note $3\sqrt{2}\times 2\sqrt{2}=6\times 2=12$. Then collect rational and surd parts.

Final answer: $-8-5\sqrt{2}$

Diagram.

$\times$	1	$-2\sqrt{2}$
4	4	$-8\sqrt{2}$
$3\sqrt{2}$	$3\sqrt{2}$	-12

Collect: $4-12=-8$; $-8\sqrt{2}+3\sqrt{2}=-5\sqrt{2}$. Note $3\sqrt{2}\times(-2\sqrt{2})=-6\times 2=-12$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 35 (a5-035)

Difficulty: Foundation | Marks: 2

Expand and simplify: $(3\sqrt{2}+\sqrt{5})(3\sqrt{2}-\sqrt{5})$

Worked solution.

Step 1: Apply the difference of two squares: $(a+b)(a-b)=a^2-b^2$.

$$(3\sqrt{2})^2-(\sqrt{5})^2$$

The brackets are conjugate pairs.

Step 2: Evaluate each square.

$$9\times 2-5=18-5=13$$

$(3\sqrt{2})^2=9\times 2=18$ and $(\sqrt{5})^2=5$.

Final answer: 13

Diagram.

$\times$	$3\sqrt{2}$	$-\sqrt{5}$
$3\sqrt{2}$	18	$-3\sqrt{10}$
$\sqrt{5}$	$3\sqrt{10}$	-5

DOTS: $(3\sqrt{2})^2=18$, $(\sqrt{5})^2=5$; cross-terms cancel, $18-5=13$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 36 (a5-036)

Difficulty: Foundation | Marks: 1

Simplify: $\sqrt{200}$

Worked solution.

Step 1: Find the largest square number that is a factor of 200.

$$\sqrt{200} = \sqrt{100 \times 2}$$

100 is the largest square factor of 200.

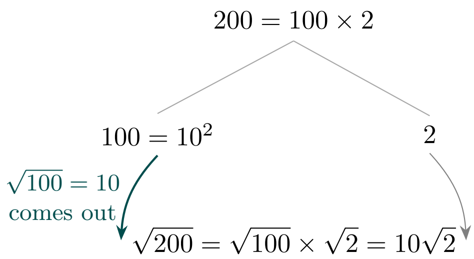
Step 2: Split and simplify.

$$\sqrt{100} \times \sqrt{2} = 10\sqrt{2}$$

$\sqrt{100} = 10$.

Final answer: $10\sqrt{2}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 37 `(a5-037)`

Difficulty: Foundation | *Marks: 1*

Simplify: $\sqrt{147}$

Worked solution.

Step 1: Find the largest square number that is a factor of 147.

$$\sqrt{147} = \sqrt{49 \times 3}$$

49 is the largest square factor of 147.

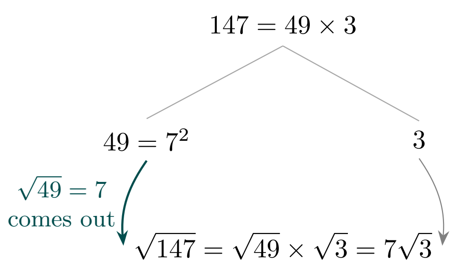
Step 2: Split and simplify.

$$\sqrt{49} \times \sqrt{3} = 7\sqrt{3}$$

$\sqrt{49} = 7$.

Final answer: $7\sqrt{3}$

Diagram.



Factor tree: split the number into its largest perfect-square factor, then the square root of that factor comes out.

Laws of Surds 38 `(a5-038)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $4\sqrt{3} \times 2\sqrt{27}$

Worked solution.

Step 1: Multiply the coefficients and the surds separately.

$$(4 \times 2) \times (\sqrt{3} \times \sqrt{27}) = 8 \times \sqrt{81}$$

$\sqrt{3} \times \sqrt{27} = \sqrt{3 \times 27} = \sqrt{81}$.

Step 2: Evaluate.

$$8 \times 9 = 72$$

$\sqrt{81} = 9$.

Final answer: 72

Diagram.

$$\boxed{m\sqrt{a} \times n\sqrt{b} = mn\sqrt{ab}}$$

$$4\sqrt{3} \times 2\sqrt{27} \xrightarrow[8\sqrt{81} = 8 \times 9]{\hspace{6em}} = \ 72$$

Surd-law card: the boxed general rule (teal) is applied to the expression.

Laws of Surds 39 `(a5-039)`

Difficulty: Foundation | *Marks: 3*

Expand and simplify: $(1 + \sqrt{6})(3 + 2\sqrt{6})$

Worked solution.

Expand and simplify

$$\begin{aligned}(1 + \sqrt{6})(3 + 2\sqrt{6}) &= 1(3 + 2\sqrt{6}) + \sqrt{6}(3 + 2\sqrt{6}) \\ &= 3 + 2\sqrt{6} + 3\sqrt{6} + 2(6) \\ &= 3 + 2\sqrt{6} + 3\sqrt{6} + 12 \\ &= 15 + 5\sqrt{6}\end{aligned}$$

Distribute each term. Note $\sqrt{6} \times 2\sqrt{6} = 2 \times 6 = 12$. Then collect rational and surd parts.

Final answer: $15 + 5\sqrt{6}$

Diagram.

$\times$	3	$2\sqrt{6}$
1	3	$2\sqrt{6}$
$\sqrt{6}$	$3\sqrt{6}$	12

Collect: $3 + 12 = 15$; $2\sqrt{6} + 3\sqrt{6} = 5\sqrt{6}$. Note $\sqrt{6} \times 2\sqrt{6} = 2 \times 6 = 12$. Area model (box method): the two binomial terms label the rows and columns; sum the cells, then collect the rational and surd parts.

Laws of Surds 40 `(a5-040)`

Difficulty: Foundation | *Marks: 3*

Triangle PQR is right-angled at Q. Side PR has length $7\sqrt{2}$ cm and side PQ has length $\sqrt{2}$ cm. Find the length of side QR in the form $k\sqrt{6}$ cm, where k is an integer.

Worked solution.

Step 1: Apply Pythagoras's theorem: $PR^2 = PQ^2 + QR^2$, so $QR^2 = PR^2 - PQ^2$.

$$QR^2 = (7\sqrt{2})^2 - (\sqrt{2})^2$$

PR is the hypotenuse (the side opposite the right angle).

Step 2: Evaluate each square.

$$QR^2 = 49 \times 2 - 1 \times 2 = 98 - 2 = 96$$

$(7\sqrt{2})^2 = 49 \times 2 = 98$ and $(\sqrt{2})^2 = 2$.

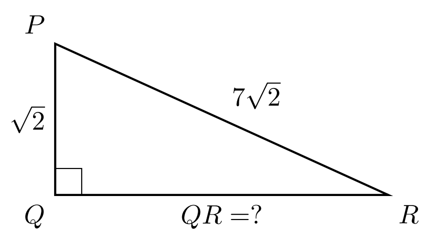
Step 3: Take the square root and simplify.

$$QR = \sqrt{96} = \sqrt{16 \times 6} = 4\sqrt{6}$$

16 is the largest square factor of 96, and $\sqrt{16} = 4$.

Final answer: $4\sqrt{6}$ cm

Diagram.



By Pythagoras, $QR^2 = PR^2 - PQ^2 = (7\sqrt{2})^2 - (\sqrt{2})^2 = 98 - 2 = 96$, so $QR = \sqrt{96} = 4\sqrt{6}$ cm.
